Basic Principles of Medicine 2
Module: ER
Lecture 04

Lecture Title:

Synthesis of Amino Acid Derivative Hormones

Dr. Sharmila Upadhya
Email: shupadhya@sgu.edu

Acknowledgements: Dr. Clayton Taylor
ctaylor2@sgu.edu

Synthesis of amino acid derivative hormones

SOM.MK.II.BPM2.1.ER.2.BCHM.1010	Discuss the biosynthesis and degradation of hormones that are derived from amino acids and analyze clinical lab tests that rely on the relevant molecule.
SOM.MK.II.BPM2.1.ER.2.BCHM.1011	Review the pathways for synthesis of the hormones derived from amino acids
SOM.MK.II.BPM2.1.ER.2.BCHM.1012	Describe the physiological role of hormones derived from amino acid precursors.
SOM.MK.II.BPM2.1.ER.2.BCHM.1013	Identify hormones formed from tyrosine and explain the biosynthesis of catecholamines.
SOM.MK.II.BPM2.1.ER.2.BCHM.1014	Highlight the use of tetrahydrobiopterin (BH4), PLP, vitamin C (ascorbic acid), and S-adenosyl methionine in the synthesis of Dopamine, Norepinephrine and Epinephrine. Identify the various enzymes required.
SOM.MK.II.BPM2.1.ER.2.BCHM.1015	Explain the role of MAO and COMT in degradation of catecholamines
SOM.MK.II.BPM2.1.ER.2.BCHM.1016	Describe VMA, metanephrine and homovanillic acid as the degradation products of the catecholamines and their clinical significance in the diagnosis of pheochromocytoma
SOM.MK.II.BPM2.1.ER.2.BCHM.1017	Explain the steps in the biosynthesis of the thyroid hormones in the thyroid gland. Identify the role of thyroglobulin and post-translational iodination of thyroid hormones to form T3 and T4.
SOM.MK.II.BPM2.1.ER.2.BCHM.1018	Outline and explain the biosynthesis of T3 and T4.
SOM.MK.II.BPM2.1.ER.2.BCHM.1019	Identify hormones and neurotransmitters formed from tryptophan.
SOM.MK.II.BPM2.1.ER.2.BCHM.1020	Indicate the steps in the formation of serotonin (5-hydroxy tryptamine; 5-HT) from tryptophan
SOM.MK.II.BPM2.1.ER.2.BCHM.1021	Explain the formation of 5-HIAA formation (Role of MAO) and analyze its clinical significance
SOM.MK.II.BPM2.1.ER.2.BCHM.1022	Indicate the precursor molecule of melatonin and identify its significance in the pineal gland
SOM.MK.II.BPM2.1.ER.2.BCHM.1023	Correlate the following disorders to the biochemical basis of the manifestations and the associated laboratory findings: Parkinson disease; Pheochromocytoma; Carcinoid syndrome.

Recommended readings

- Lippincott's Illustrated Reviews – Biochemistry; 9th Edition; [Amino Acids: Conversion to Specialized Products | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library]
- 🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
  - Section: Catecholamines; Page 333-334
  - Section: Serotonin; Page 335
  - Section: Iodine; Page 480-482 (Iodine, and thyroid hormone)

Types of Hormones

- Steroid Hormones
- Protein/peptide hormones
- **Amino acid derived hormones**
    - **Catecholamines- Dopamine, Norepinephrine, Epinephrine (derived from Phenylalanine or Tyrosine)**
    - **Serotonin- derived from Tryptophan**
    - **Melatonin- derived from Serotonin (Tryptophan)**
    - **Thyroid hormone- Thyroxine (T4) and T3 (derived from phenylalanine or Tyrosine)**

Catecholamines: Origin and function

- Dopamine and Norepinephrine: Synthesized in the brain and function as neurotransmitters
- Epinephrine: Synthesized from norepinephrine in the adrenal medulla
- Outside the CNS, norepinephrine and epinephrine are released in response to stress (fright, exercise, cold and low levels of glucose)

Catecholamine Synthesis: Enzymes and coenzymes

Phenylalanine

↓ Tetrahydrobiopterin (BH4) | **Phenylalanine Hydroxylase**

Tyrosine

↓ Tetrahydrobiopterin (BH4) | **Tyrosine Hydroxylase**

DOPA (Dihydroxyphenylalanine) (L-DOPA)

↓ PLP (Vit B6) | **DOPA Decarboxylase**

Dopamine

↓ Vitamin C | **Dopamine β-Hydroxylase**

Norepinephrine

↓ S-Adenosyl methionine (SAM) | **Phenylethanolamine N-methyl transferase (PNMT)**

Epinephrine

SOM.MK.II.BPM2.1.ER.2.BCHM.1011

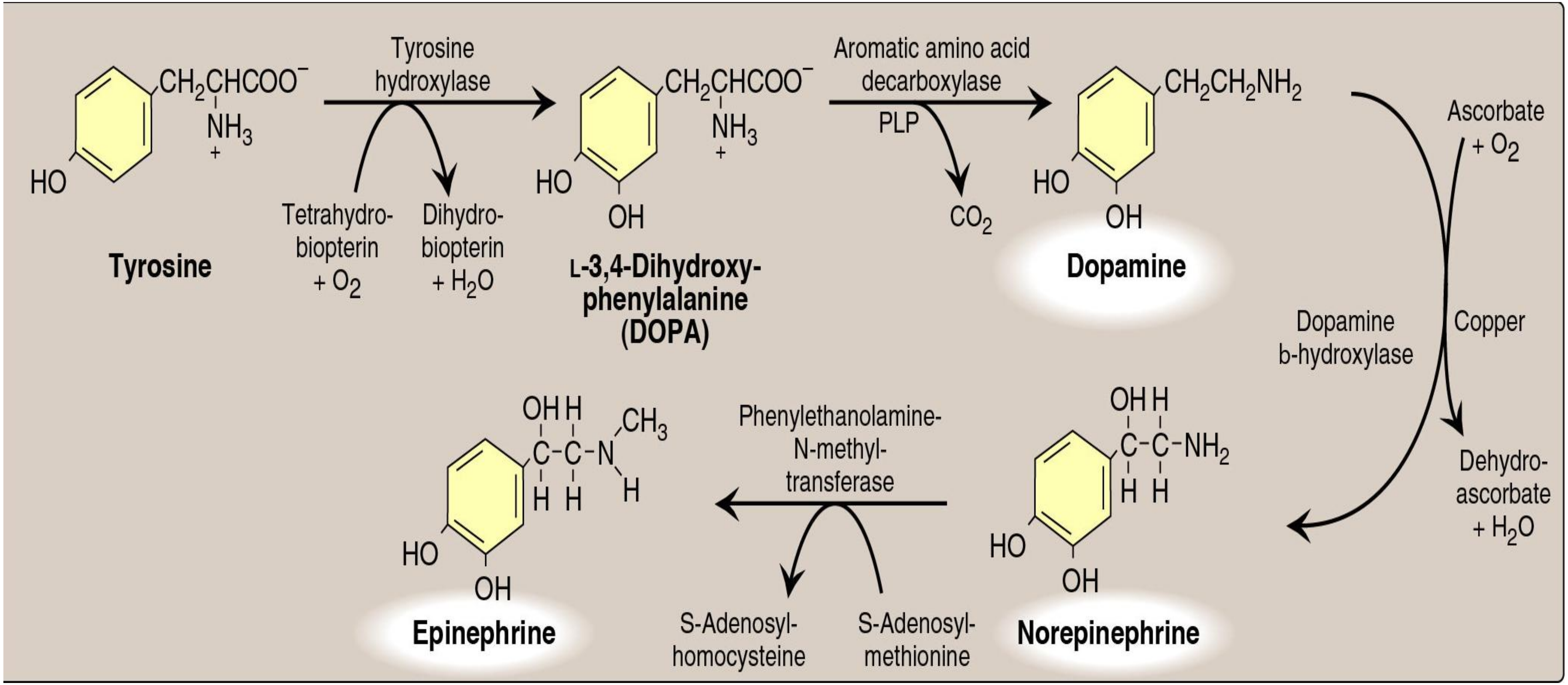
Catecholamine Synthesis

- **Phenylalanine** is converted to tyrosine by phenylalanine hydroxylase (BH4 as coenzyme)
- Phenylalanine is dietary essential, however in the absence of phenylalanine in the diet, tyrosine becomes essential.
- **Tyrosine** is hydroxylated to **L-DOPA**
- Dopa decarboxylase (**PLP/ vitamin B6**) forms Dopamine
- Dopamine beta-hydroxylase forms Norepinephrine (**requires vitamin C**)
- PNMT forms epinephrine (requires S-adenosyl methionine as methyl donor)

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

SOM.MK.II.BPM2.1.ER.2.BCHM.1014

Catecholamine Synthesis

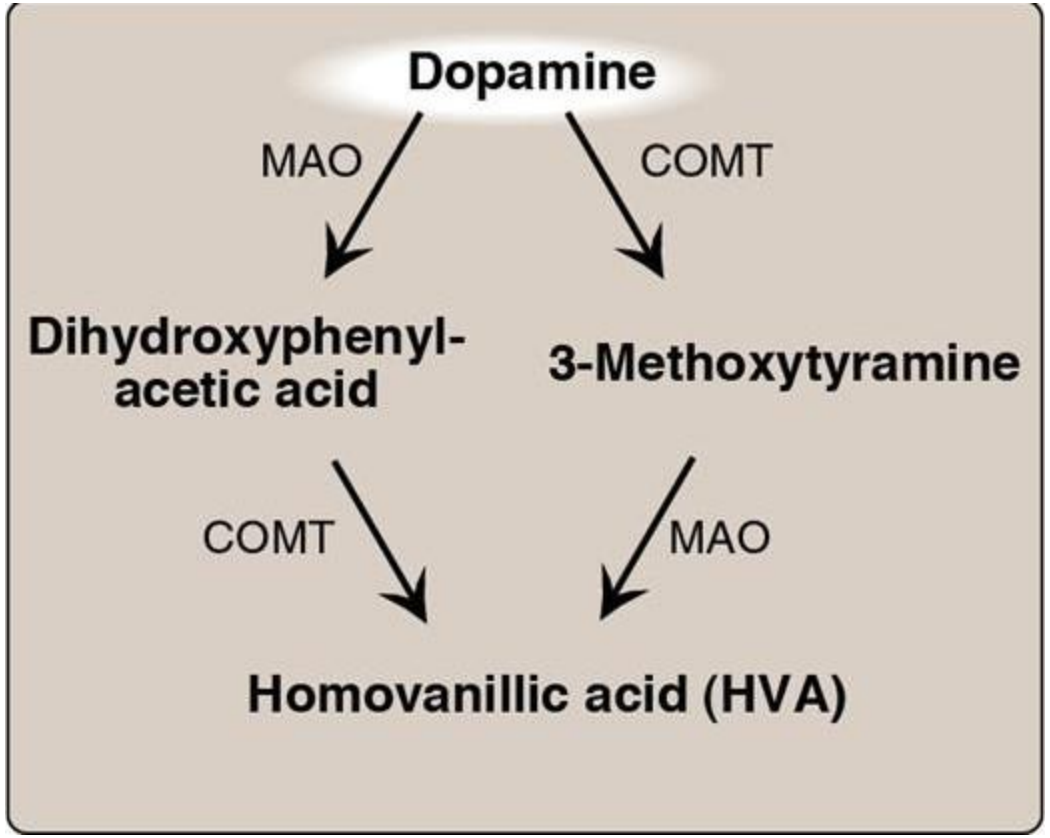


SOM.MK.II.BPM2.1.ER.2.BCHM.1011

Catecholamine: Dopamine

- Important neurotransmitter in basal ganglia of brain
- Dopamine **inhibits Prolactin secretion**
- Types of Dopamine receptors: D1, D2, D3, D4 and D5
- Dopamine receptors in brain (all are G-proteins): Gs, Gi or Gq
  - Gs – Activates adenylate cyclase → Increases cAMP
  - Gi – Inhibits adenylate cyclase → Decreases cAMP
  - Gq – Activates phospholipase C → Forms DAG and IP3

Dopamine Degradation

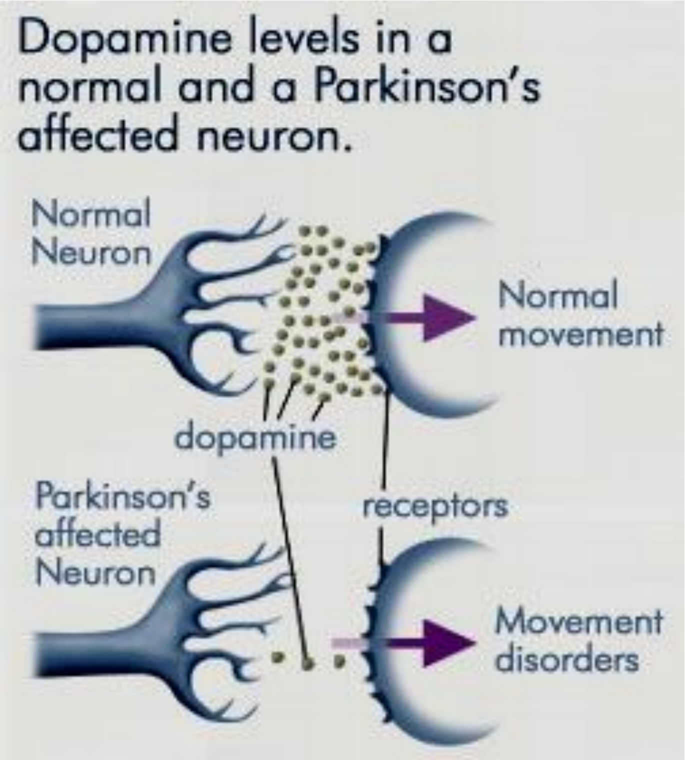


- Monoamine oxidase (MAO)
- Catechol-O-methyltransferase (COMT)
- Reactions can occur in either order
- **Dopamine → Homovanillic acid (Urine)**

SOM.MK.II.BPM2.1.ER.2.BCHM.1015

SOM.MK.II.BPM2.1.ER.2.BCHM.1016

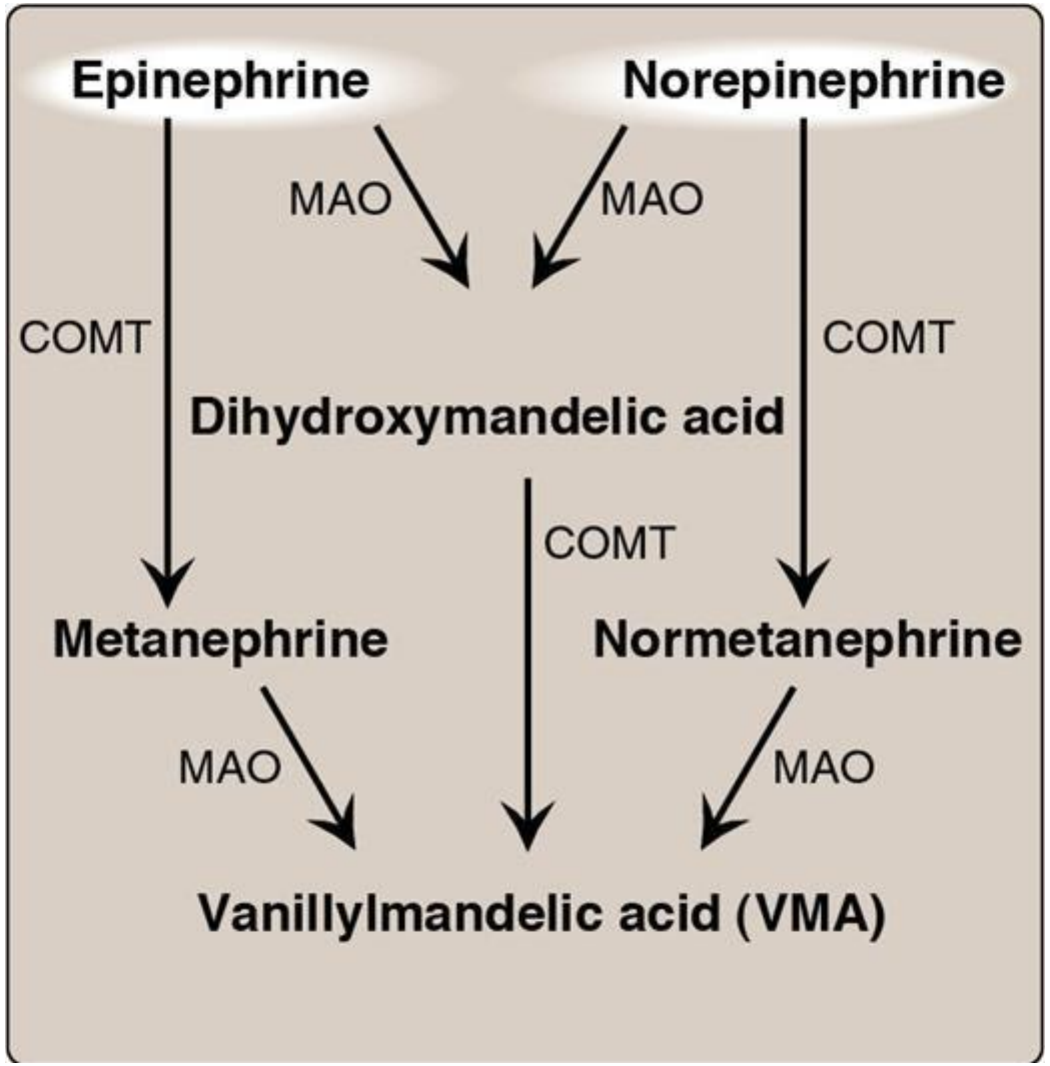
Parkinson disease



- Complex progressive neurological disorder involving movement and higher cognitive function.
- Degeneration of dopamine producing neurons in substantia nigra (basal ganglia of brain).
- Reduced dopamine (neurotransmitter).
- Triad: Tremor (Pill-rolling resting tremors), Cog-wheel rigidity and Bradykinesia.
- Other cognitive and behavioral symptoms include be memory loss, mood disturbances, postural instability.

- Treatment: L-DOPA
- L-DOPA → Dopamine in brain

Epinephrine and norepinephrine Degradation



- Monoamine oxidase (MAO)
- Catechol-O-methyltransferase (COMT)

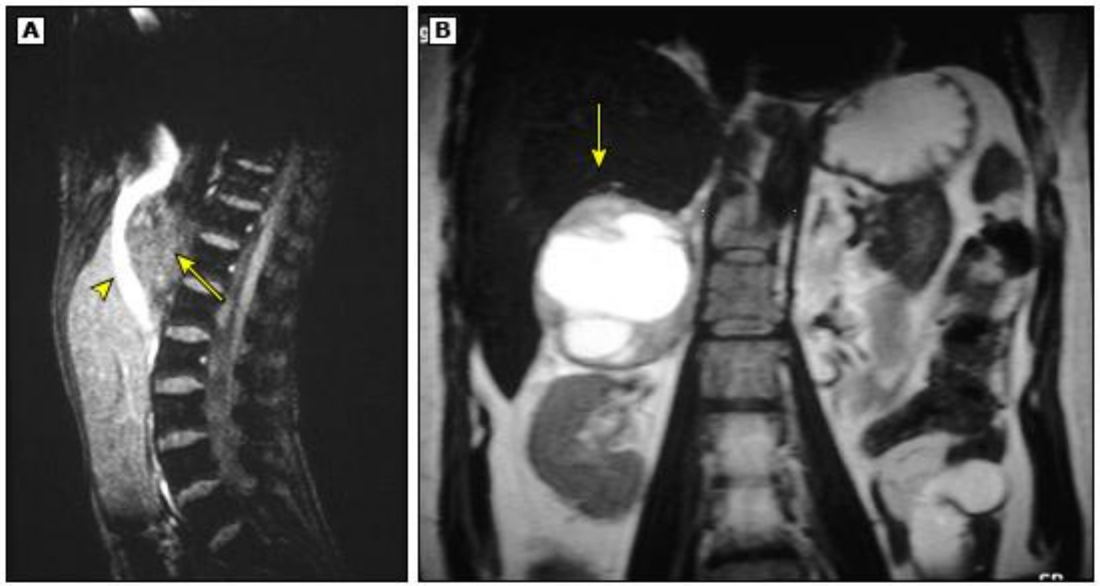
- Reactions can occur in either order

- **From Epinephrine and Norepinephrine→ Vanillylmandelic acid (Urine)**

SOM.MK.II.BPM2.1.ER.2.BCHM.1015

SOM.MK.II.BPM2.1.ER.2.BCHM.1016

Pheochromocytoma



UpToDate: MRI findings in adrenal pheochromocytomas.
(A) Sagittal MRI demonstrates a mass (arrow) displacing the inferior vena cava anteriorly (arrowhead). This caval displacement is typically seen with tumors arising in the adrenal gland.
(B) Coronal MRI of the abdomen demonstrates a large mass in the right adrenal bed (arrow), lying immediately superior to the kidney. The low signal intensity in the center is caused by hemorrhage into the tumor.

- Tumor of **adrenal medulla**
- Derived from chromaffin cells (embryological development from neural crest)
- Triad of Pheochromocytoma: Headache, Sweating, Tachycardia
- **Episodic symptoms:** Due to overproduction of **epinephrine** by tumor
- Hypertension during episode
- Diagnosis: 24-hour urinary measurement during episode
- High urinary VMA, metanephrine, and catecholamines are diagnostic

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

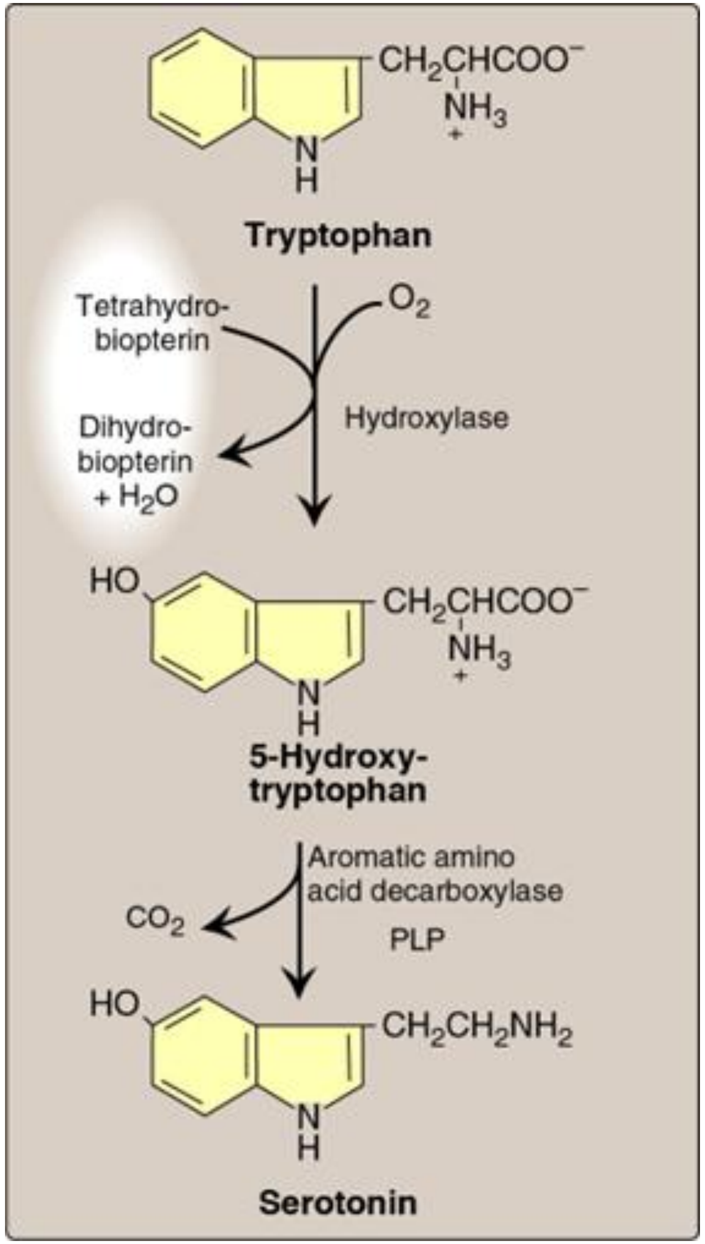
SOM.MK.II.BPM2.1.ER.2.BCHM.1016

SOM.MK.II.BPM2.1.ER.2.BCHM.1023

Serotonin: Origin and function

- Also called 5-hydroxytryptamine (5-HT)
- Synthesized in intestinal mucosa
- Released by platelets
- CNS: neurotransmitter
- Physiologic roles: pain perception, sleep regulation, appetite, temperature, blood pressure, cognitive functions and **mood**.

Serotonin synthesis



- **Tryptophan** converted to **5-hydroxy-tryptophan** by a **BH4 hydroxylation** reaction.
- **5-hydroxytryptophan** is converted to **Serotonin** by **aromatic amino acid decarboxylase** (**PLP** coenzyme)
- Serotonin converted to **melatonin**

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

SOM.MK.II.BPM2.1.ER.2.BCHM.1020

Serotonin (5-hydroxy tryptamine) degradation

- Serotonin metabolized to **5-hydroxyindole acetic acid (5-HIAA)** by **MAO**

Tryptophan

Tetrahydrobiopterin BH4

Tryptophan hydroxylase

5 hydroxy tryptophan

Pyridoxal phosphate (PLP; Vit B6)

Amino acid Decarboxylase

5-hydroxy tryptamine (Serotonin)

Monoamine oxidase (MAO)

5-HIAA excretion

Some modifications

Melatonin

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

SOM.MK.II.BPM2.1.ER.2.BCHM.1021

Carcinoid Syndrome

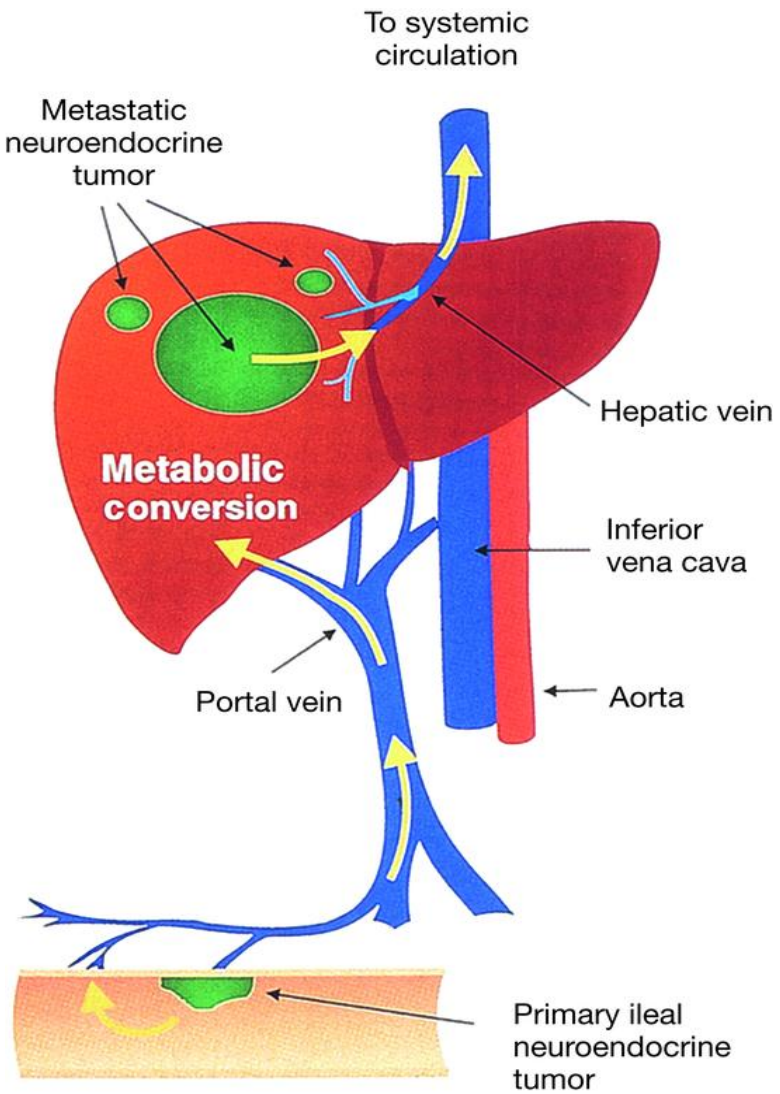
- **Gastrointestinal: Neuroendocrine tumors (carcinoids)** mainly overproduce serotonin.
- More than 95% of serotonin is secreted in the GI tract by enterochromaffin cells (Neuroendocrine cells).
- **Symptoms**: Flushing, diarrhea, bronchospasm, and cardiac tricuspid valve disease.
- **Lab tests**: **Elevated urinary 5-HIAA**

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

SOM.MK.II.BPM2.1.ER.2.BCHM.1023

Carcinoid Syndrome

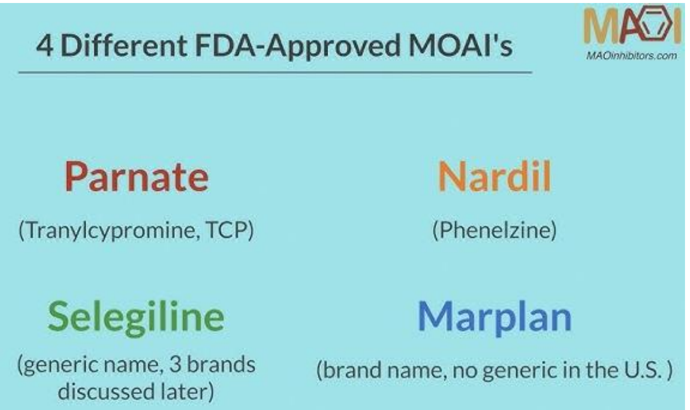
- These symptoms arise depending on the location of the tumor
- When tumor metastasizes to liver, bone, lung, and ovary, then serotonin is secreted (escapes) directly into the systemic circulation



SOM.MK.II.BPM2.1.ER.2.BCHM.1023

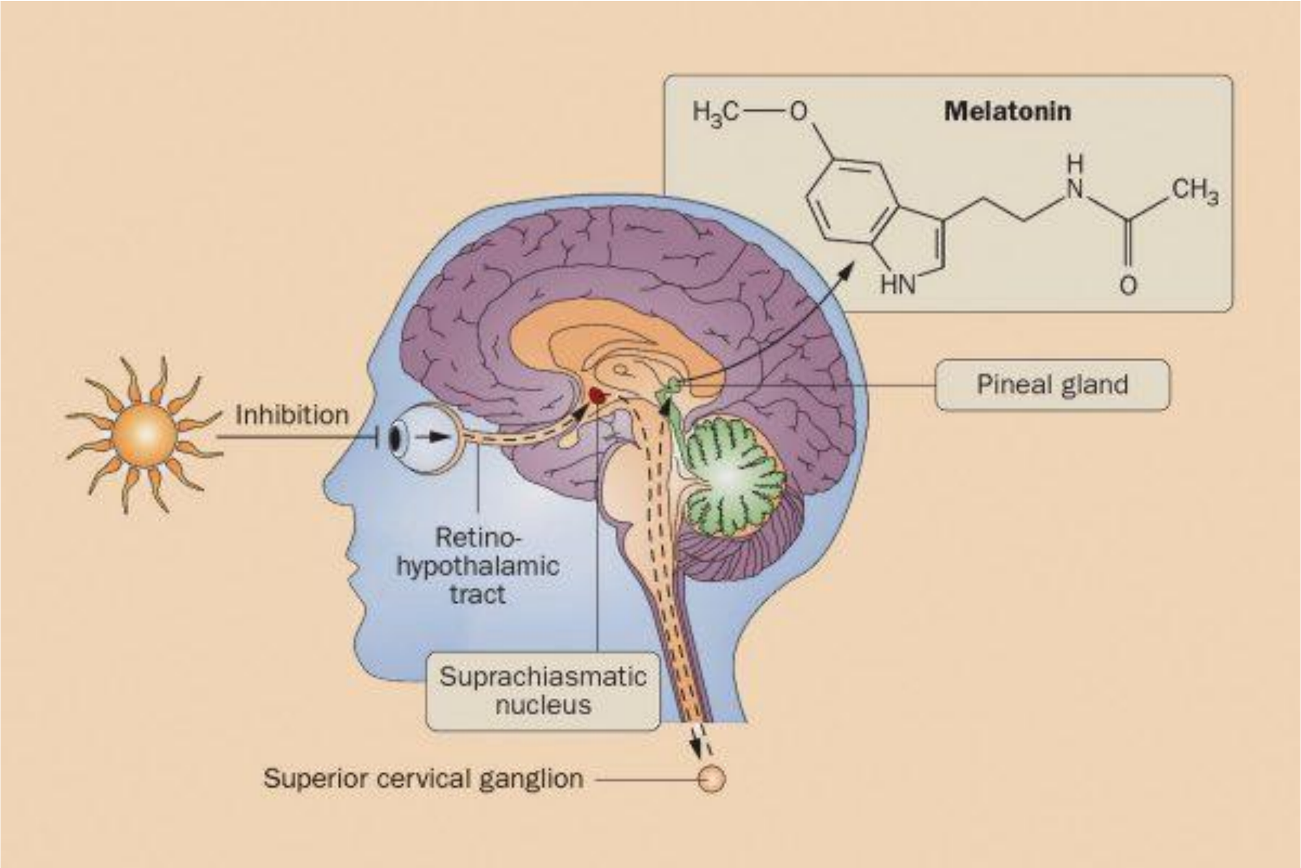
Pharmacological applications of MAO and COMT

- Serotonin modulates appetite, mood, and sexual function.
- Serotonin degraded to **5-hydroxyindole acetic acid (5-HIAA)** by **Monoamine Oxidase (MAO)**.
- **MAO inhibitors (MAOI)** increase serotonin levels in the CNS
  - A class of antidepressant drugs works via serotonin
  - Slows down serotonin degradation
- **Selective serotonin reuptake inhibitors (SSRI)**
  - Different class of drug that also functions via serotonin
  - SSRI increases serotonin levels in synapses



Melatonin synthesis and function

- Hormone from **pineal gland**
- From **tryptophan**
- Regulates **circadian rhythm**
- **Regulates sleep**
  - Exposure to light inhibits melatonin
  - Less melatonin disrupts circadian rhythm
- Endogenous antioxidant: free radical scavenger



SOM.MK.II.BPM2.1.ER.2.BCHM.1012

SOM.MK.II.BPM2.1.ER.2.BCHM.1022

Seasonal affective disorder

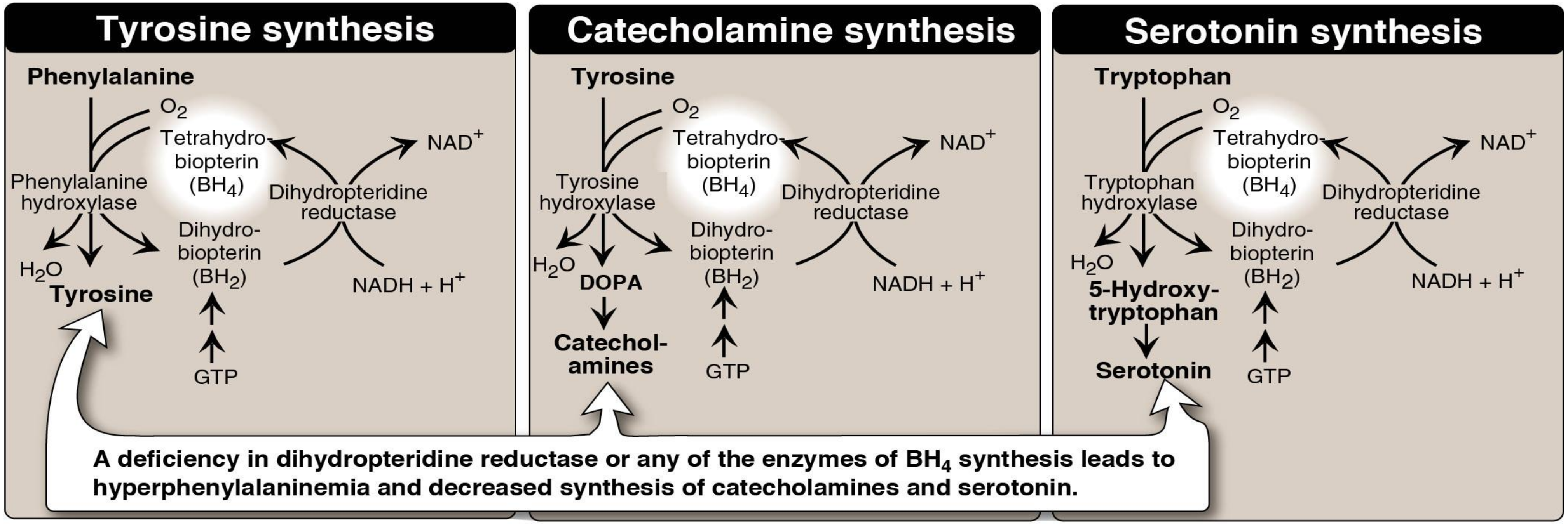
Patients with **SAD ("winter depression"): Light therapy**

- **Shorter days**
- **Less daylight**
- **Chemical changes causing depression**

SAD light Therapy decreases melatonin production which can lead to improved symptoms.

SOM.MK.II.BPM2.1.ER.2.BCHM.1022

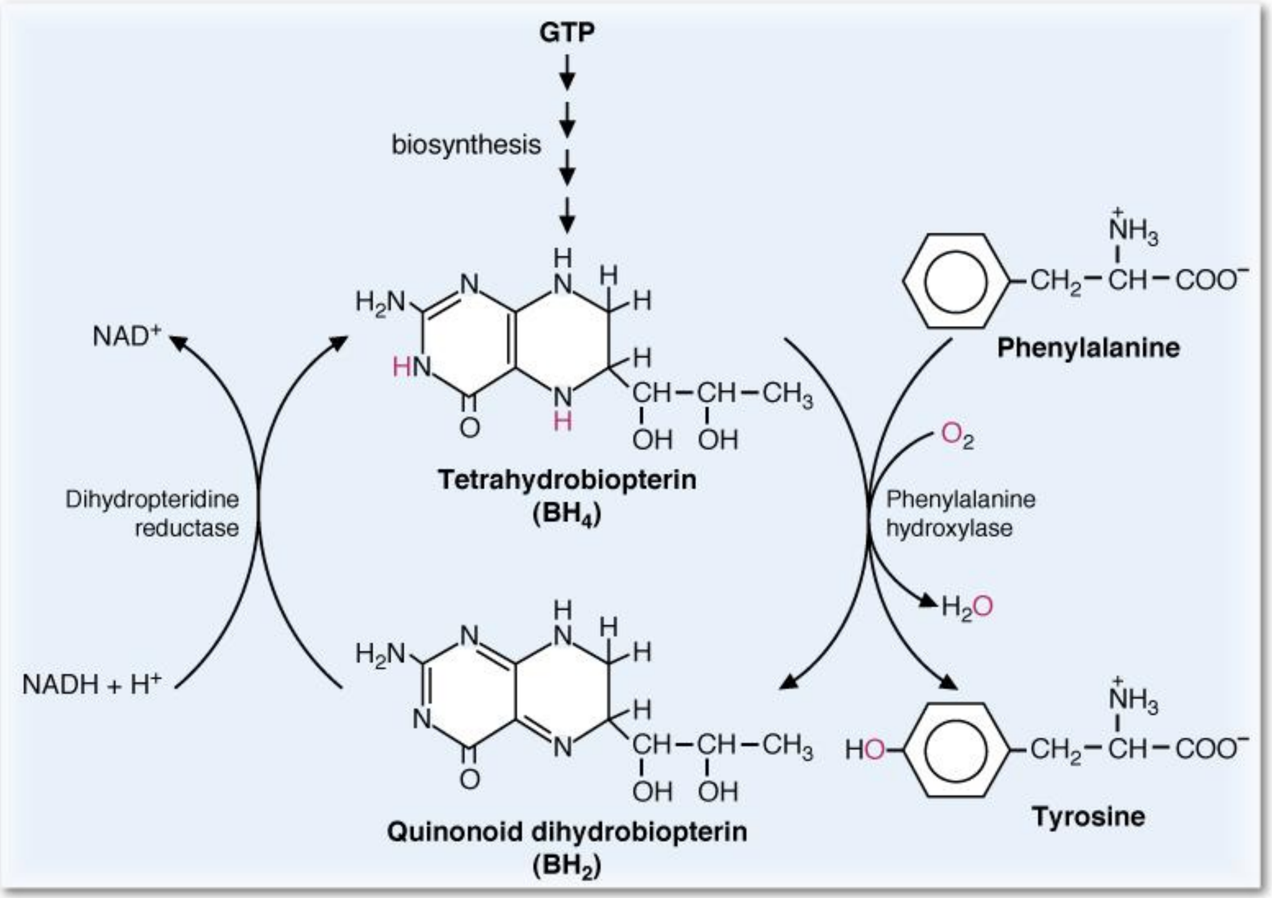
Tetrahydrobiopterin (BH4)



- Coenzyme for **amino acid hydroxylation** reactions
- Reactions requiring tetrahydrobiopterin:
  1. **Phenylalanine hydroxylase (converts Phe to Tyr)**
  2. **Tyrosine hydroxylase (converts Tyr to DOPA)**
  3. **Tryptophan hydroxylase (converts Trp to 5-hydroxy tryptophan)**

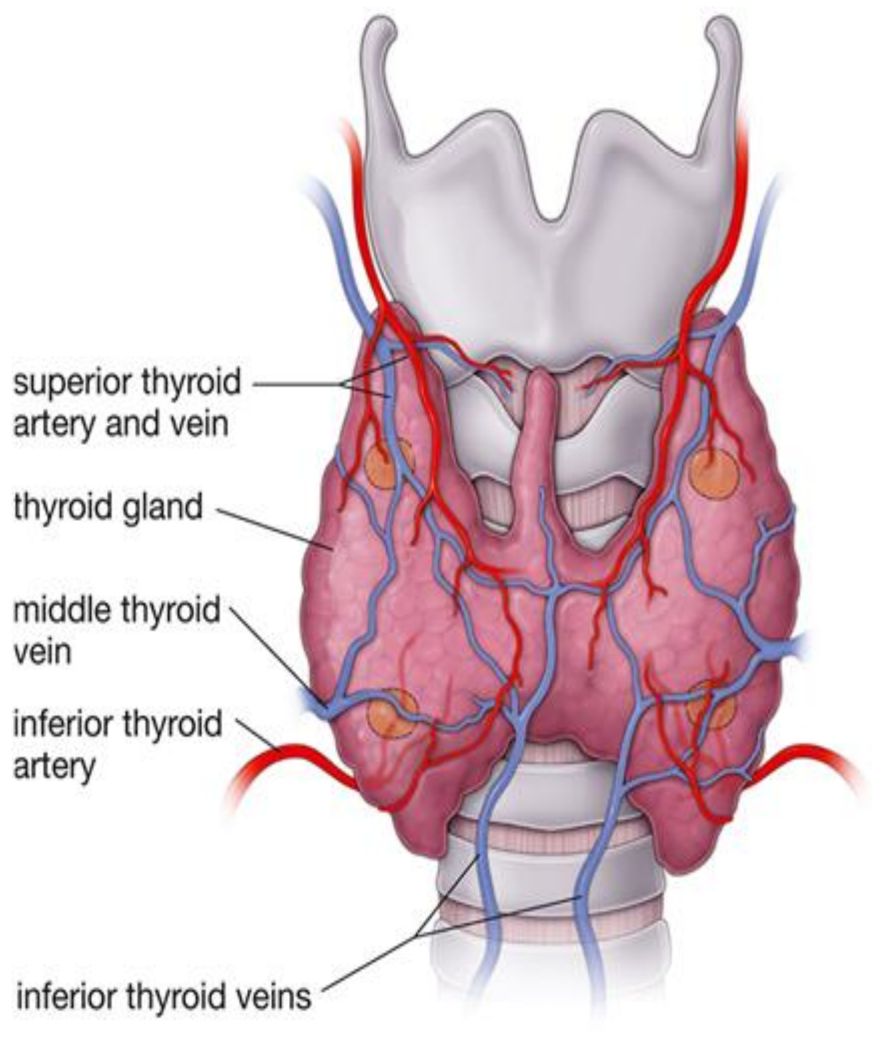
SOM.MK.II.BPM2.1.ER.2.BCHM.1014

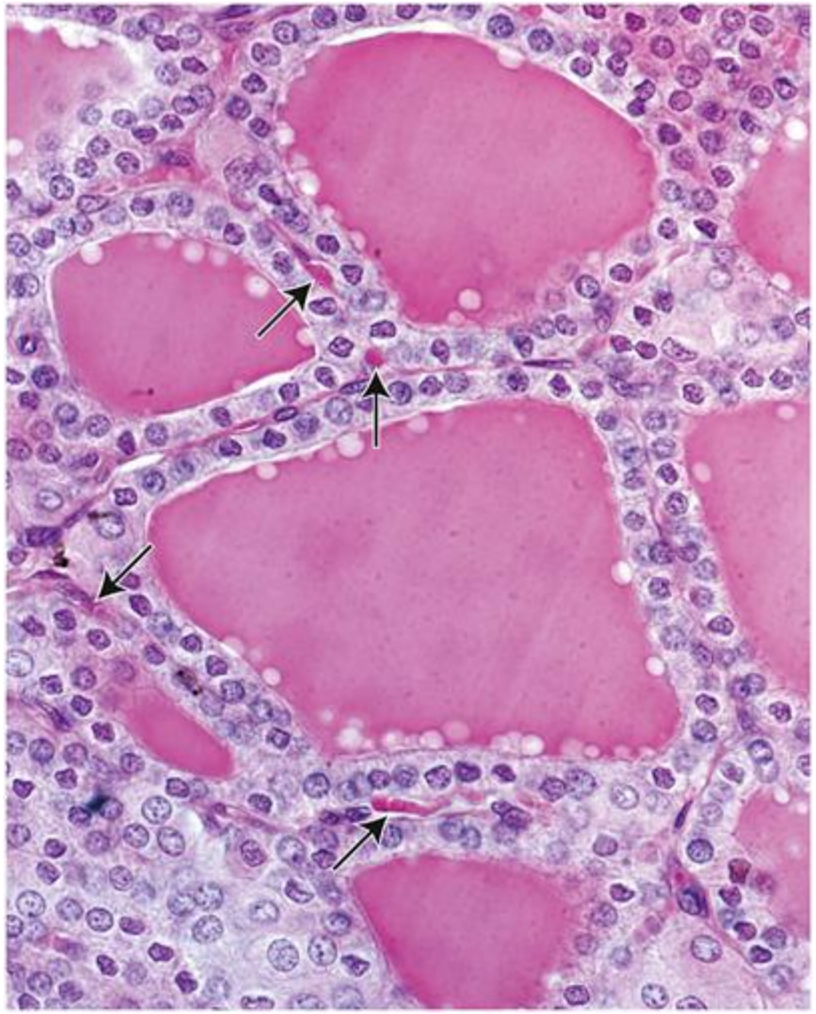
Tetrahydrobiopterin deficiency



- Deficiency of **tetrahydrobiopterin** results in hyperphenylalaninemia.
- Decreased synthesis of neurotransmitters
- Results in **severe intellectual disability and seizures**

Thyroid hormone: Origin and Mechanism of action





- **Synthesis: Uses Iodine**
- **Occurs in Thyroid gland.**
- **Regulated by: TSH**

- **Mechanism of action of thyroid hormones: Bind to intracellular receptors and regulates gene expression**

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

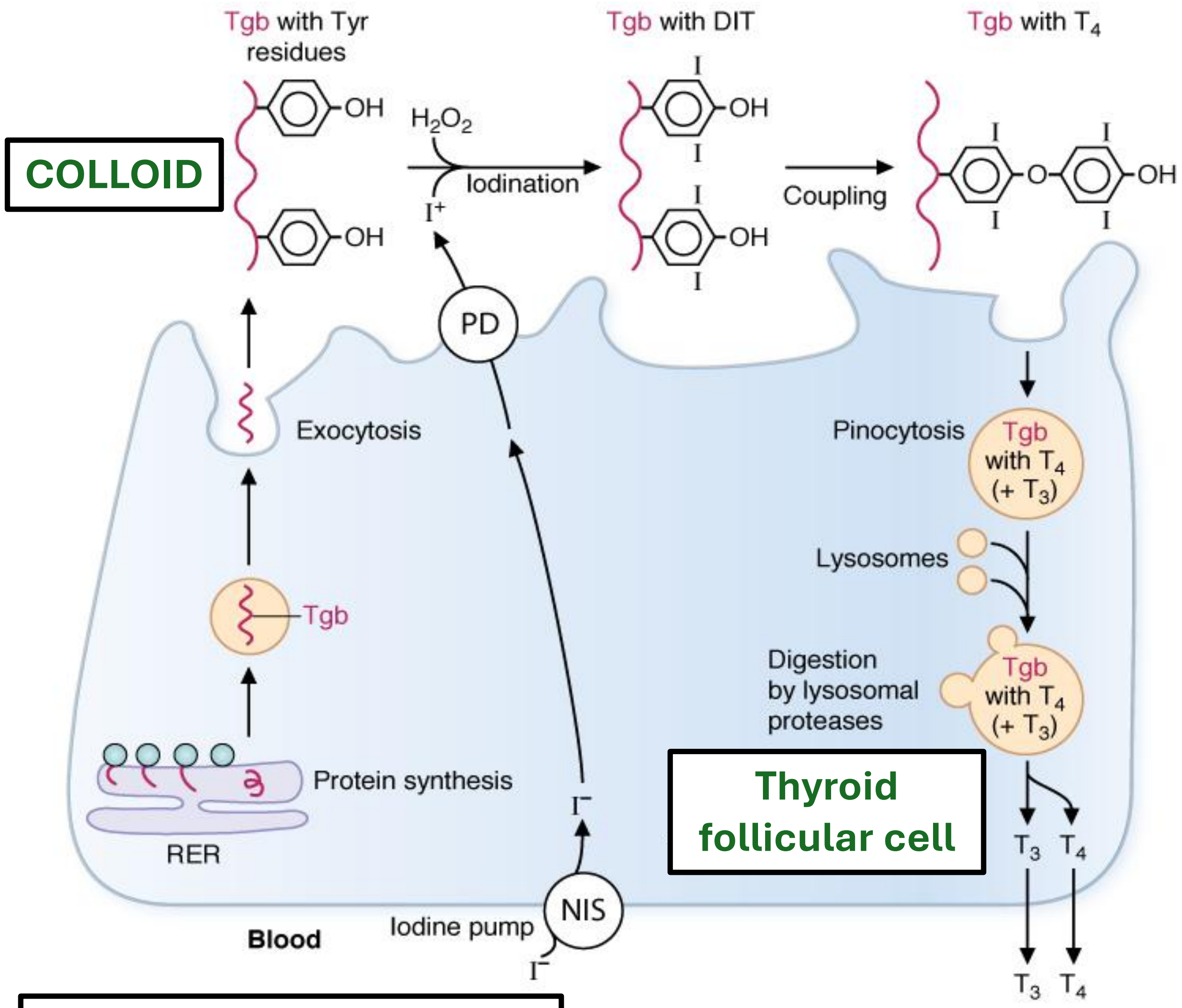
Steps of thyroid hormone synthesis

1. Trapping of iodide
2. Oxidation of iodide (thyroperoxidase)
3. Iodination of thyroglobulin (thyroperoxidase)
4. Coupling of MIT and DIT (thyroperoxidase)
5. Release of T3 and T4
6. Formation of T3 from T4 (5'-deiodinase)
7. Inactivation of T4 (5-deiodinase)

Regulated by **Thyroid stimulating hormone (TSH)**

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

1. Trapping of iodide



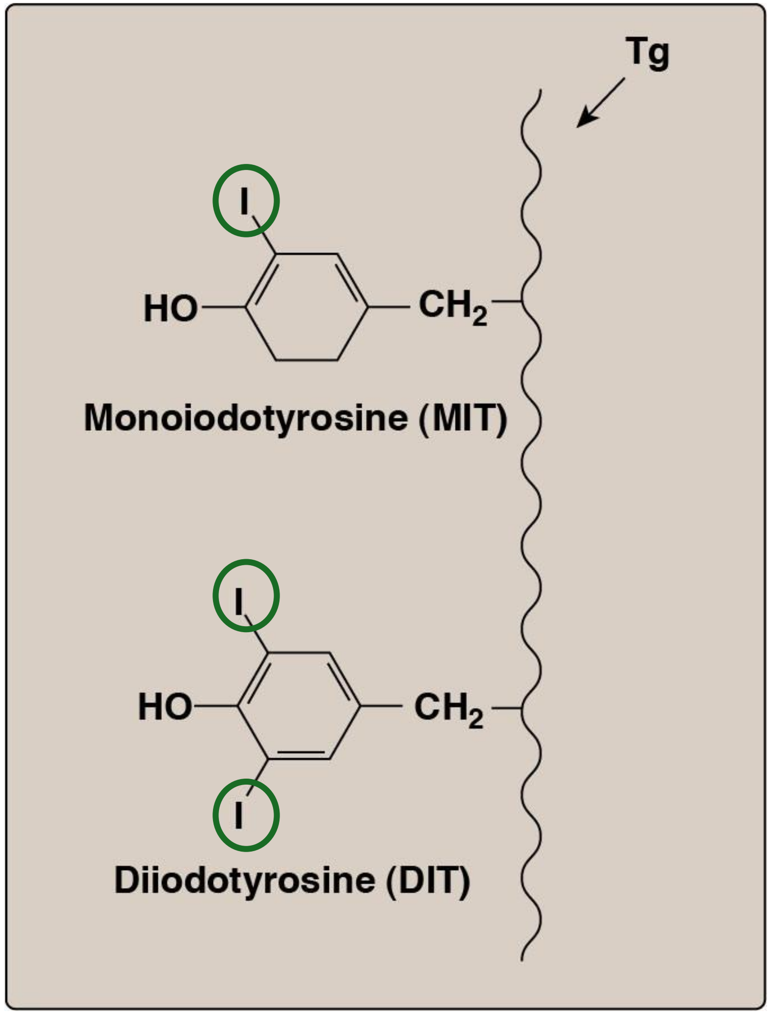
SOM.MK.II.BPM2.1.ER.2.BCHM.1017

Trapping: Circulating iodide (I^-) is taken up (trapped) and concentrated in the epithelial follicular cells of the thyroid gland by **Sodium-Iodide cotransporters**.

Pendrin: transports I^- from FOLLICULAR CELL into colloid

Thyroglobulin (Tgb) protein rich in tyrosine residues

Thyroperoxidase: 2. Oxidation of iodide and 3. Iodination of thyroglobulin



Thyroperoxidase (TPO):

- **Oxidation** of iodine ions I^- to I_2.
- Uses I_2 to **iodinate** tyrosine residues of thyroglobulin to form **monoiodothyrosine (MIT) and diiodotyrosine (DIT)**

SOM.MK.II.BPM2.1.ER.2.BCHM.1013

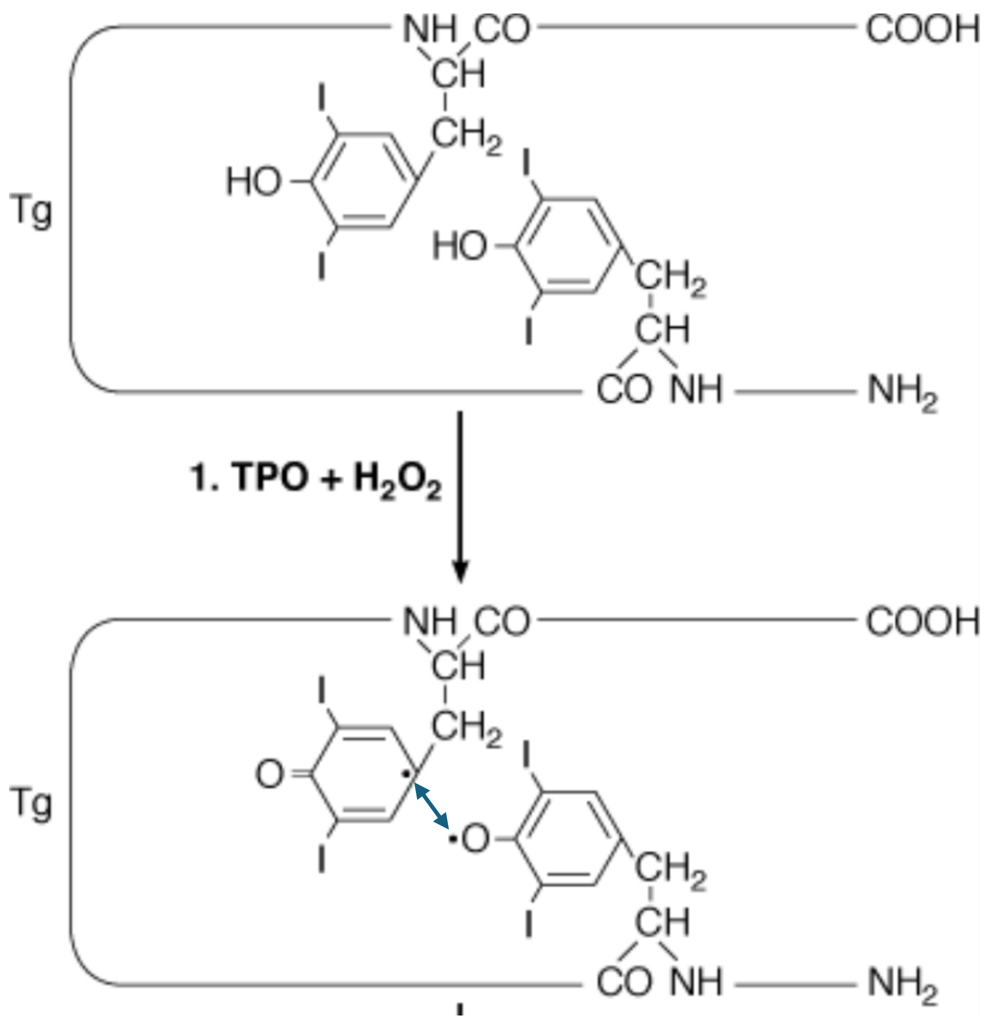
SOM.MK.II.BPM2.1.ER.2.BCHM.1017

Thyroperoxidase: 4. Coupling of MIT and DIT

Thyroperoxidase (TPO) then acts as peroxidase **(Coupling)**

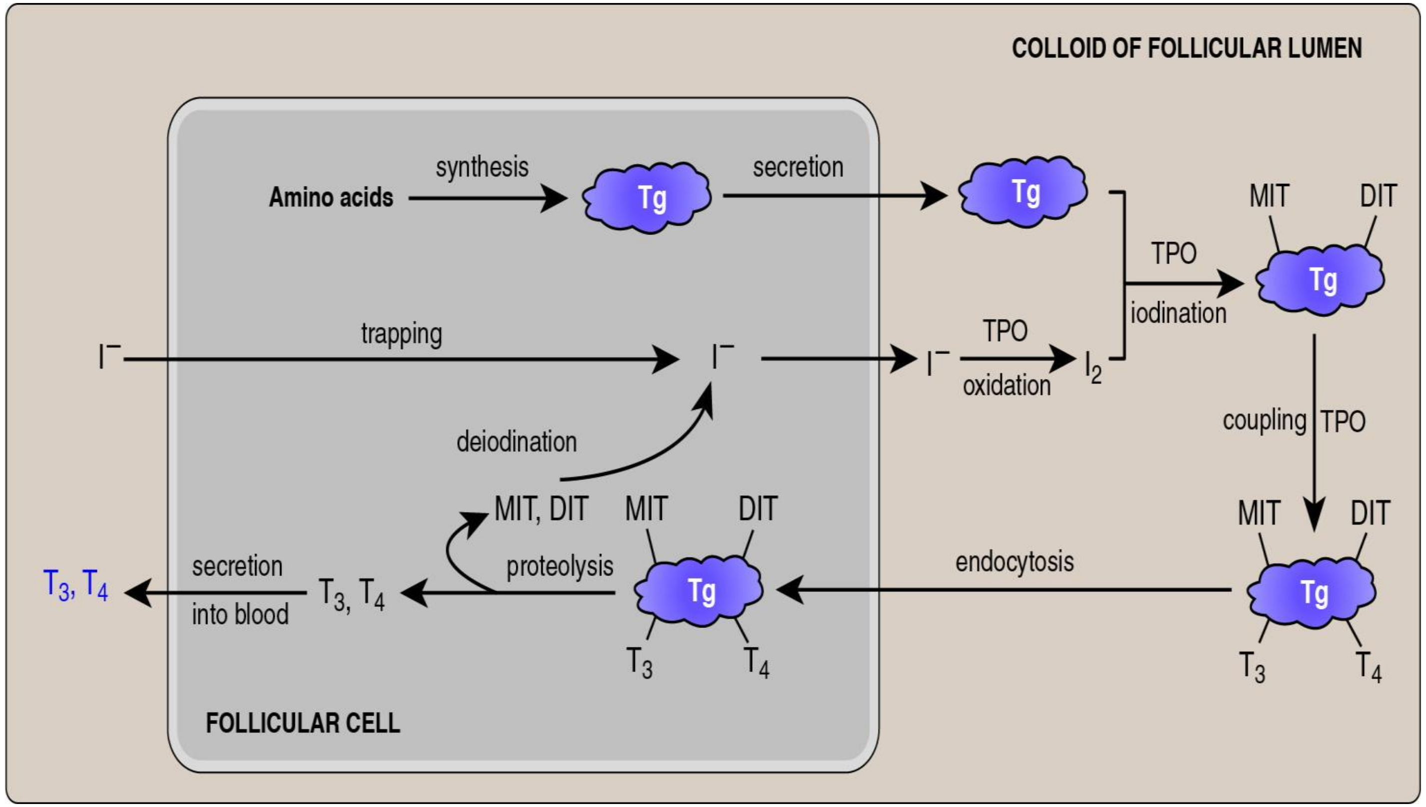
Links two iodotyrosine side chains

- **2 DIT → T4 (Thyroxine)**
- **MIT +DIT → T3 (triiodothyronine)**



SOM.MK.II.BPM2.1.ER.2.BCHM.1017

Thyroid hormone synthesis (Summary and 5. Release)



- **Trapping**
- **Oxidation of I^-**
- **Iodination**
- **Coupling**
- **Release**

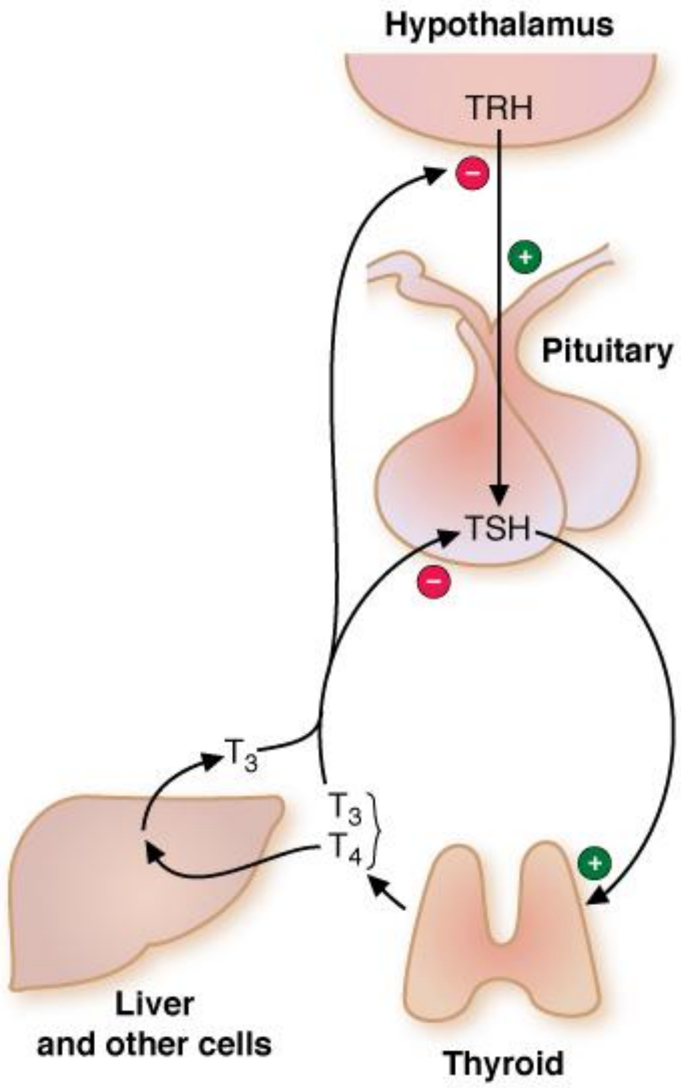
Lysosomes: Degrade thyroglobulin and release T3 and T4.

80-90% of secreted thyroid hormone is T4

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

SOM.MK.II.BPM2.1.ER.2.BCHM.1018

Feedback Regulation of Thyroid hormone levels



Thyroid releasing hormone (TRH) from **hypothalamus** stimulates release of TSH from the anterior pituitary gland

Thyroid stimulating hormone (TSH) stimulates synthesis of T3 and T4 from **thyroid**.

T3/T4 inhibits TSH by negative feedback

SOM.MK.II.BPM2.1.ER.2.BCHM.1018

Thyroid hormone synthesis: Formation of T3

Activation: T4 converted to active **T3** by enzyme **5'-deiodinase** (note the **prime**) **"five prime deiodinase"**

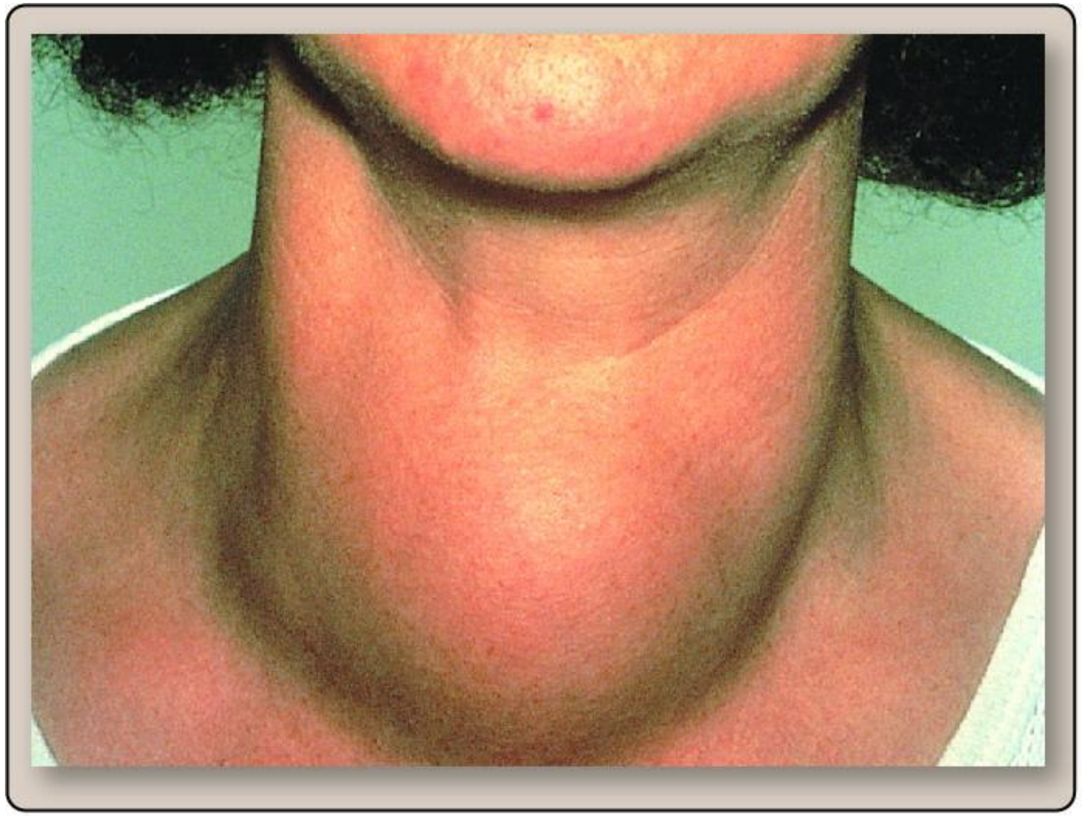
- In thyroid gland, or in target tissues

Inactivation: T4 inactivated by conversion to ***reverse* T3 by 5-Deiodinase** (note this is different than five-prime deiodinase enzyme)

Enzyme	Pronunciation	Function
5'-deiodinase	5-prime deiodinase	Converts T4 to **T3** (more **active** form)
5-deiodinase	5-deiodinase	Converts T4 to reverse T3 (inactivate form)

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

Iodine deficiency



- **Low ingestion of iodine (I)** can result in goiter due of overstimulation of TSH on thyroid gland because of underproduction of T3 and T4.
- **Hypothyroidism**- several symptoms of severe hypothyroidism are fatigue, weight gain, decreased thermogenesis and decreased metabolic rate.
- **Congenital Hypothyroidism-** hormone deficiency during fetal and infant development. **Irreversible intellectual disability**, hypothermia, **hearing loss** and **short stature**.

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

Graves' disease

- **Graves' disease** caused by antibodies that mimic **TSH** which results in **overproduction of T3 and T4.**
- Graves' disease is most common cause of **hyperthyroidism** which is caused by overproduction of T3 and T4.
- **Hyperthyroidism**- nervousness, weight loss, increased perspiration, heart rate and protruding eyes.

SOM.MK.II.BPM2.1.ER.2.BCHM.1018

Thank you